

GRAYSCALE IMAGE REPRESENTATION

Practical Limitations, Analysis & Communication

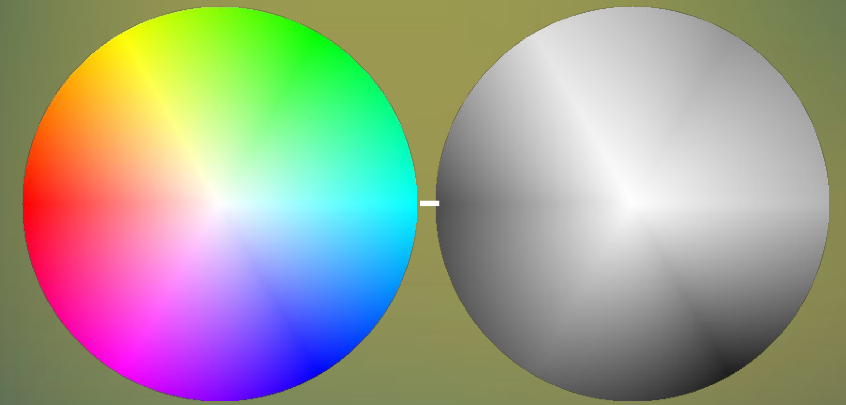
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colour → grayscale



What Is Grayscale Representation?

- Each pixel stores a single intensity value instead of separate red, green, and blue channels.
- Standard depth is 8 bits per pixel — 256 levels, ranging from 0 (black) to 255 (white).
- Widely used for compression, printing, medical scans, and classic computer-vision pipelines.



Colour → Gray Conversion



$$Y = 0.299R + 0.587G + 0.114B$$

Luminance formula: $Y = 0.299R + 0.587G + 0.114B$
— red, green, and blue combine into one brightness value.

Practical Limitations



Loss of Colour Information

All hue and saturation data is discarded — only brightness remains.



Metamerism (Colour Confusion)

Distinct colours with equal luminance become indistinguishable gray tones.



Quantization / Banding Artifacts

Low bit-depth reduces smooth gradients to visible stepped bands.



Unsuitable for Colour-Critical Tasks

Traffic-light detection, skin-tone analysis, and fruit ripeness checks fail.



Analyzing Banding: Bit-Depth Reduction

8-bit gradient — 256 levels



3-bit gradient — 8 levels (posterized)



Zoomed-in pixel values (a real 6×6 patch)

169	152	129	129	151	173
131	110	101	100	125	144
100	78	69	64	79	97
112	95	56	74	87	109
145	124	105	90	124	125
171	137	116	116	141	162

Each cell is one pixel's stored gray value (0–255) — this is the literal number a computer sees.

Reading the Comparison

- Top strip: smooth transition, no visible steps — typical camera output.
- Middle strip: same data forced into just 8 gray values — hard edges appear between bands.
- Human vision resolves far more than 8 levels, so banding is immediately visible.
- Fix: dithering, or raising bit-depth to 10/12-bit before display or print.



Analyzing Colour Loss on a Real Scene

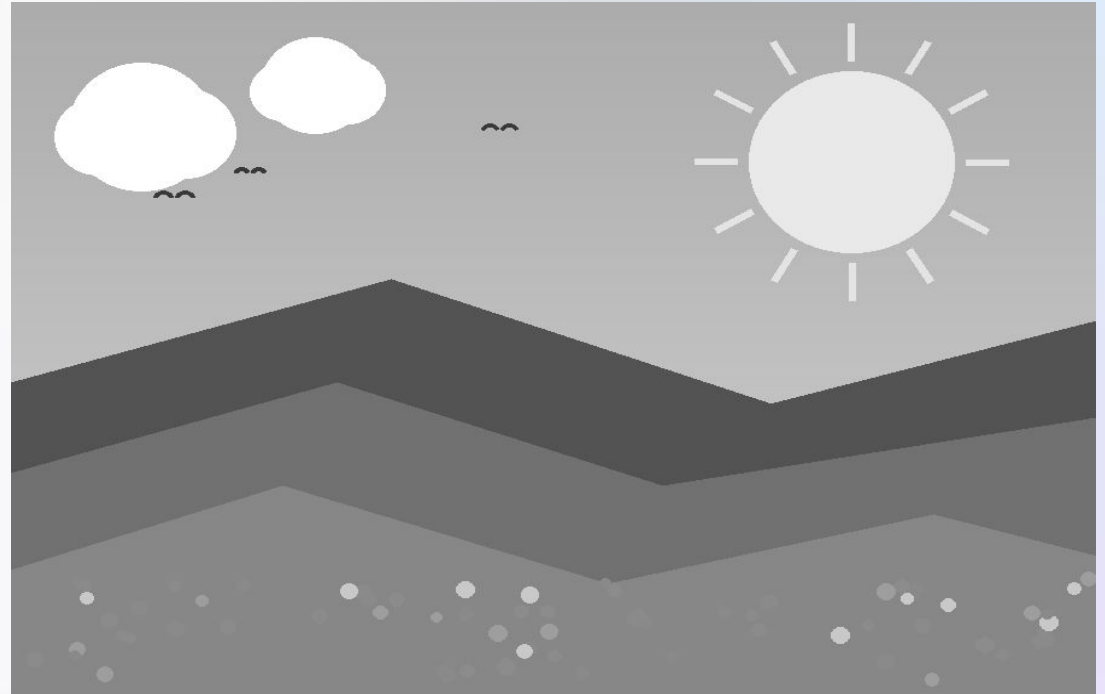
Original



Same idea, isolated: three very different colours — all reduce to almost the same gray.



Grayscale



Orange (250,128,0), green (0,255,0), and neutral gray (150,150,150) all map to luminance ≈ 150 .



Key Takeaways

- Grayscale representation trades colour for simplicity — one intensity value per pixel, ideal for speed and storage.
- Its core limitations — lost colour data, metamerism, and quantization banding — are measurable, not just theoretical.
- Simple Python tools (NumPy, Pillow) are enough to reproduce and demonstrate every limitation shown here.
- Choose grayscale for efficiency; choose colour when hue distinctions are essential to the task.

Thank You

GEOTAG PHOTO

